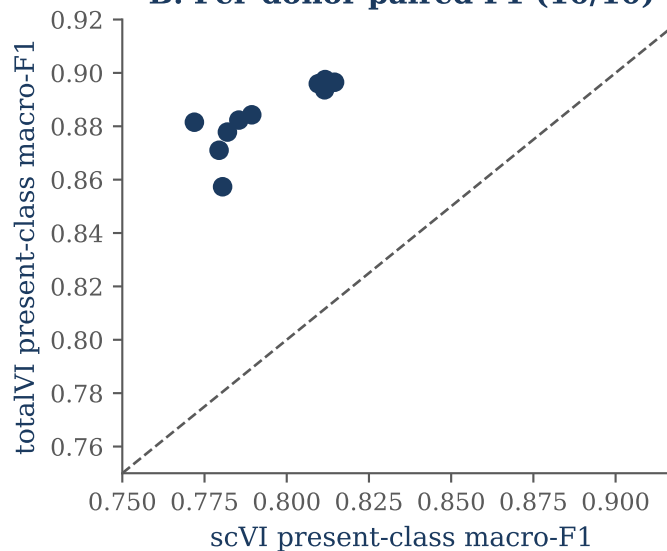


A. Lawlor design

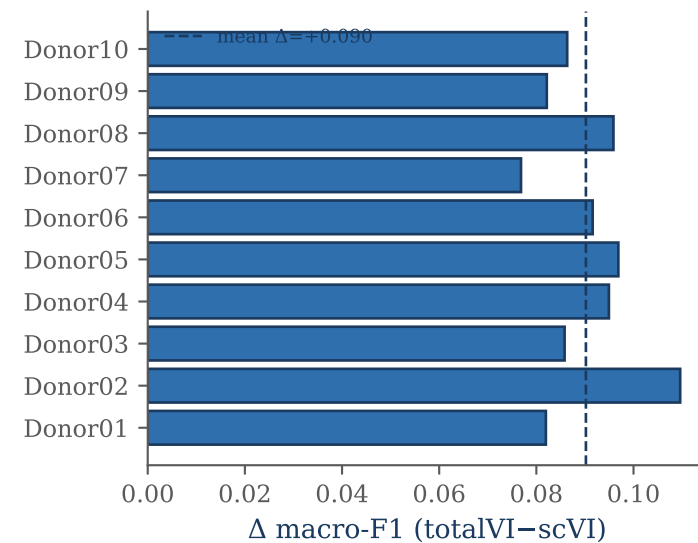
10 biological donors
5 donor folds
10 seeds / fold
100 deep-model runs
50 matched fold×seed pairs

Primary biological unit: DONOR
Fold×seed pairs $\neq$ biological n

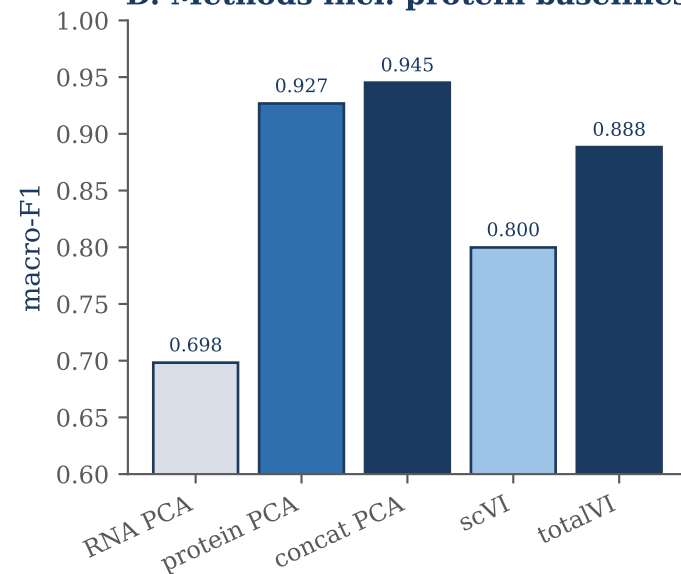
B. Per-donor paired F1 (10/10)



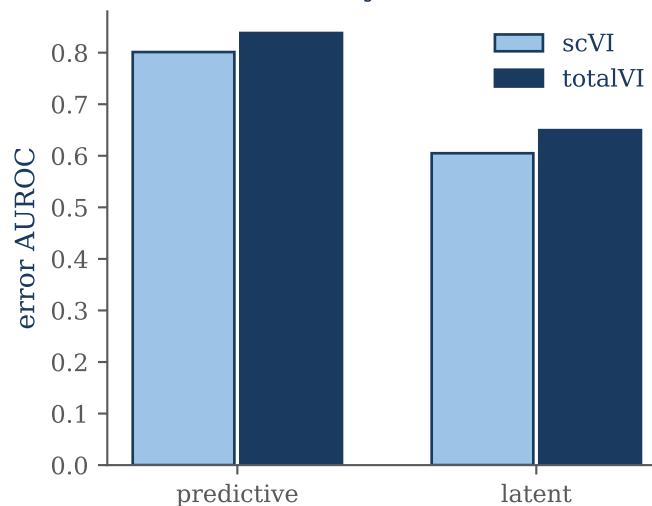
C. Per-donor Δ F1



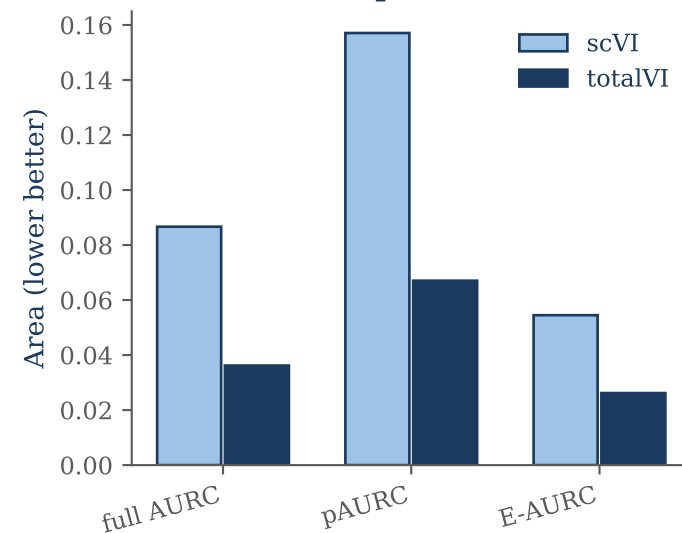
D. Methods incl. protein baselines



E. Uncertainty discrimination



F. Selective prediction areas



Note: Lawlor author labels are protein-gated; strong protein PCA / concat performance indicates label-modality alignment, not universal architecture superiority.